Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2002-I-Q05] | Part A (Q1-Q5)

How many distinct planar squares have at least two vertices among the vertices of a regular dodecagon?

平面中有多少个不同正方形，其至少两个顶点属于某正十二边形的顶点集?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2004-I-Q03] | Part A (Q1-Q5)

A convex polyhedron has 26 vertices, 60 edges, 24 triangular faces, and 12 quadrilateral faces. How many vertex-joining segments have endpoints neither adjacent nor on any common face?

凸多面体有二十六顶点、六十棱、二十四个三角面及十二个四边形面。连接两顶点且两点不相邻、也不在同一面上的线段有多少条?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2007-II-Q05] | Part A (Q1-Q5)

Count unit lattice squares whose interiors lie entirely in the first quadrant and strictly below the line $9x + 223y = 2007$.

求内部完全位于第一象限且位于直线 $9x + 223y = 2007$ 下方的单位格正方形数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2014-I-Q05] | Part A (Q1-Q5)

For the twelve vertices of a regular dodecagon, count subsets Q for which some circle has all of Q strictly inside and all remaining vertices strictly outside. Include the empty subset.

对正十二边形的十二顶点，求子集 Q 的个数，使存在一圆将 Q 全部严格包含于内、其余顶点全部严格留于外，包括空集。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2016-I-Q03] | Part A (Q1-Q5)

A regular icosahedron is oriented with top vertex T, bottom vertex B, and horizontal upper and lower five-vertex rings. Count simple edge paths from T to B that never move upward. Each ring is a cycle, each upper-ring vertex has two lower-ring neighbors, T joins all upper vertices, and B joins all lower vertices.

正二十面体以一顶点 T 朝上、对顶点 B 朝下，中间为水平的上、下两个五顶点环。求从 T 到 B、沿棱且从不向上、也不重复顶点的路径数。每个环为五边形循环，上环每个顶点连接下环两个顶点，T, B 分别连接各自邻近环的全部顶点。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2022-II-Q05] | Part A (Q1-Q5)

On a circle, 20 distinct points are labeled 1 through 20 in circular order. Join two points precisely when the absolute difference of their labels is prime. Count triangles whose three vertices are labeled points and whose three edges are drawn.

圆上 20 个不同点按顺序标为 1 至 20。两点标号差的绝对值为质数时连线。求三个顶点均为已标点、三条边均已画出的三角形数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2025-II-Q03] | Part A (Q1-Q5)

Color all twelve unit edges of a 2×2 square grid red or blue so each unit square has two edges of each color. Count the colorings.

将 2×2 方格的十二条单位边染红或蓝，使每个小正方形恰有两红边、两蓝边。求染色数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2008-II-Q10] | Part B (Q6-Q10)

On the sixteen lattice points $\{0,1,2,3\}^2$, a growing path visits distinct points with strictly increasing successive distances. If the maximum point count is m and exactly r paths attain it, find mr.

在十六个格点 $\{0,1,2,3\}^2$ 上，路径访问互异点且相邻距离严格递增。若最多可有 m 个点，达到者有 r 条，求 mr。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

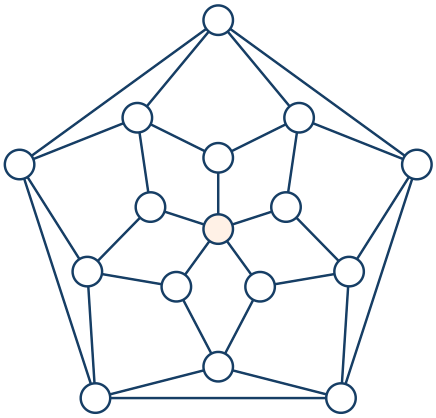
Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2012-I-Q07] | Part B (Q6-Q10)

Sixteen students stand at the vertices of the connected undirected graph shown. They share 3360 coins. Simultaneously each sends all their coins equally among their neighbors, after which every student has the original number again. Find the initial coin count at the central vertex.

十六名学生位于图示连通无向网络的顶点，共有 3360 枚硬币。所有人同时将自己的硬币平均送给各邻点学生，交换后每人硬币数与原来相同。求中心学生原有硬币数。



Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2018-I-Q07] | Part B (Q6-Q10)

A right prism has unit-side regular hexagonal bases and height two. Count isosceles triangles determined by three of its twelve vertices, including equilateral triangles.

直六棱柱底面为单位边长正六边形，高为二。求十二顶点任取三点构成的等腰三角形数，包括等边三角形。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2022-II-Q09] | Part B (Q6-Q10)

Two parallel lines contain respectively 7 and 5 marked points. Draw every segment between points on different lines, along with both full lines. Assume every interior crossing involves exactly two drawn segments. How many bounded planar regions result?

两条平行线上分别有 7 个和 5 个标记点，画出所有跨两线的连线段以及两条整直线。假设两线之间每个交点恰由两条线段相交形成。求有界区域数目。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2012-II-Q14] | Part C (Q11-Q15)

Nine labeled people each shake hands with exactly two others, and no pair shakes twice. Count the possible undirected handshaking patterns modulo 1000.

九名不同的人各与另外两人握手，每对至多握一次。求不同无向握手关系的数目模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q11] | Part C (Q11-Q15)

Each pair of five labeled towns is joined by one road. Count orientations of all ten roads for which every town can reach every other by a directed route.

五个有标签城镇中，每两镇之间恰有一条道路。将十条路都改为单向，求使任意一镇都能沿有向路线到达任意另一镇的定向数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q13] | Part C (Q11-Q15)

Let $f(n)$ count three-vertex subsets of a regular n-gon forming isosceles triangles, including equilateral ones. Find the sum of all integers $n \geq 3$ for which $f(n+1) - f(n) = 78$.

令 $f(n)$ 为正 n 边形顶点中任选三点组成等腰三角形的个数，等边也计入。求满足 $f(n+1) - f(n) = 78$ 的所有整数 $n \geq 3$ 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2017-II-Q14] | Part C (Q11-Q15)

In the three-dimensional grid $\{1,\ldots,10\}^3$, count distinct straight lines containing exactly eight grid points.

在三维点阵 $\{1,\ldots,10\}^3$ 中，求恰含八个点阵点的不同直线数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2023-I-Q14] | Part C (Q11-Q15)

Two distinguishable clock hands start at 12. Each move advances exactly one hand clockwise by one hour mark. Count length-144 move sequences that visit all 12^2 ordered hand positions once before returning to the start; report the count modulo 1000.

两根可区分的钟表指针从 12 出发，每步只使其中一根顺时针前进一个刻度。求经过全部 12^2 种有序位置各一次、并在第 144 步返回起点的移动序列数模 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2024-II-Q15] | Part C (Q11-Q15)

Inside a fixed regular dodecagon, count all rectangles whose four sides lie along sides or diagonals of the dodecagon. Rectangle vertices may be chord intersections.

在固定正十二边形内部，求四条边均位于其边或对角线上的矩形个数，矩形顶点可以是弦的交点。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2025-II-Q11] | Part C (Q11-Q15)

How many perfect matchings of the vertices of a regular 24-gon use twelve segments all of the same length?

正二十四边形的二十四个顶点两两配对，画十二条等长线段，每个顶点恰使用一次。求画法数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 4.6 Graphs & geometric counting

图论与几何计数

[AIME-2026-II-Q15] | Part C (Q11-Q15)

Count ordered 7-tuples $a_i \in \{1, 2, 3\}$ such that both $\sum_i a_i$ and $\sum_{i=1}^{7} a_i a_{i+1} a_{i+3}$ are divisible by 3, with indices interpreted cyclically modulo seven.

求有序七元组 $a_i \in \{1, 2, 3\}$ 的个数，使 $\sum_i a_i$ 与 $\sum_{i=1}^{7} a_i a_{i+1} a_{i+3}$ 均被三整除，下标按模七循环。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程